

ELK Stack Detailed Troubleshooting Runbook

Environment

Ubuntu EC2, Filebeat 7.17.29, Logstash 7.17.29, Elasticsearch 7.17.29, Kibana 7.17.29.

Problem Summary

Kibana initially returned HTTP 500. Filebeat had YAML issues, application JAR was corrupt, some Grok parsing failed, Kibana saved-object indices disappeared, and data views required recreation.

Step 1 - Verify Elasticsearch

Command:

```
curl http://localhost:9200/_cluster/health?pretty
```

Observed:

cluster_name=my-cluster

status=yellow

number_of_nodes=1

active_primary_shards=5/6 during troubleshooting

Yellow was expected because replica shards cannot be allocated on a single-node cluster.

Step 2 - List Indices

Command:

```
curl http://localhost:9200/_cat/indices?v
```

Observed during session:

.geoip_databases

logs-2026.07.03 (6 documents)

read_me

Later, after Kibana recovery, only system indices existed temporarily because the logs index had been removed and recreated.

Step 3 - Verify Documents

Command:

```
curl 'http://localhost:9200/logs-2026.07.03/_search?pretty'
```

Observed log fields included @timestamp, log_level=INFO, log_message='This is a Filebeat test', host.name=ip-10-11-1-100, cloud.region=ap-south-1, instance id=i-097b283f1cf3cc5a9. Some events contained tag _grokparsefailure for unmatched formats.

Step 4 - Kibana Failure

Browser returned:

```
{"statusCode":500,"error":"Internal Server Error","message":"An internal server error occurred."}
```

Step 5 - Kibana Logs

journalctl showed:

Saved object index alias [.kibana_7.17.29] not found

This indicated missing Kibana saved-object indices.

Step 6 - Verify .kibana

Commands:

```
curl '_cat/indices/.kibana?v'  
curl '_cat/aliases/.kibana?v'
```

Initially only .kibana-event-log existed. No .kibana alias existed.

Step 7 - Restart Kibana

Commands:

```
sudo systemctl restart kibana  
sudo journalctl -u kibana -f
```

After restart the indices .kibana_7.17.29_001 and .kibana_task_manager_7.17.29_001 were recreated and Kibana UI opened successfully.

Step 8 - Data View

Created data view logs-* with time field @timestamp. Discover initially showed no data until time range and data view were corrected.

Step 9 - Dashboard

Created Lens visualization using logs-* and verified donut chart showing 100% of documents grouped by @version.keyword, cloud.instance.id.keyword=i-097b283f1cf3cc5a9 and cloud.availability_zone.keyword=ap-south-1a.

Useful Commands

```
filebeat test config  
filebeat test output  
systemctl status filebeat/logstash/elasticsearch/kibana  
journalctl -u SERVICE -f  
curl _cat/indices?v  
curl _cluster/health?pretty  
curl _search?pretty
```